

# Data Science Report

## Data Cleaning

| Column Name          | Situation                       | Action                          | Justification                                                                                                                                                                                 |
|----------------------|---------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brand                | Satisfactory                    | Merged                          | These will only be for labelling laptops, not comparison, so they were combined.                                                                                                              |
| Model                |                                 |                                 |                                                                                                                                                                                               |
| Screen Size          | Stored as non-numerical         | Made numerical                  | Numerical data allows numerical analysis.                                                                                                                                                     |
| Hard Disk            |                                 |                                 |                                                                                                                                                                                               |
| RAM                  |                                 |                                 |                                                                                                                                                                                               |
| Price                |                                 |                                 |                                                                                                                                                                                               |
| Colour               | Irrelevant                      | Dropped                         | Colour is irrelevant for a laptop's usability and recommendation, it's not a functional consideration.                                                                                        |
| Special Features     | 34% Missing values, listed data |                                 | Missing data causes unreliability. The available information often lists basic features, which might be present in laptops not explicitly listing them, biasing analysis.                     |
| CPU Speed            | 87% Missing values              |                                 | Missing data causes unreliability. Relevant CPU information can be derived from the CPU column for more reliable comparisons.                                                                 |
| Rating               | 60% Missing values              |                                 | Data is largely missing; ratings, being subjective, are prone to bias. Questionable reliability due to the unknown number of ratings per entry.                                               |
| CPU                  | Excessive levels                | Grouped together similar levels | Values usually belong to a series with similar properties, grouping simplifies comparison. Acknowledging rarely occurring values in a group recognizes additional options, giving them value. |
| OS                   |                                 |                                 |                                                                                                                                                                                               |
| Graphics             |                                 |                                 |                                                                                                                                                                                               |
| Graphics Coprocessor |                                 |                                 |                                                                                                                                                                                               |
| Hard Disk            | Inconsistent format             | Standardised units              | Numerical data is stored with a uniform unit for consistency.                                                                                                                                 |
| RAM                  |                                 |                                 |                                                                                                                                                                                               |
| Model                |                                 | Lowercased, removed whitespace  | Easier to compare categorical data with a consistent format.                                                                                                                                  |
| CPU                  |                                 |                                 |                                                                                                                                                                                               |
| OS                   |                                 |                                 |                                                                                                                                                                                               |

|                      |                                             |                                                  |                                                                                                                                                                                                     |
|----------------------|---------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Graphics             |                                             |                                                  |                                                                                                                                                                                                     |
| Graphics Coprocessor |                                             |                                                  |                                                                                                                                                                                                     |
| Screen Size          | Missing values                              | Mode-filled missing values                       | With a small standard deviation and missing value count, the high-frequency mode suggests missing values may align with it with minimal impact. Figures 1 & 2 highlight this.                       |
| RAM                  |                                             |                                                  |                                                                                                                                                                                                     |
| Graphics             | Conflicts with Graphics Coprocessor         | Repositioned some values to Graphics Coprocessor | Now indicates whether graphics are integrated or dedicated, most data was already in this form, other values exist in Graphics Coprocessor. This information is also important for recommendations. |
| Graphics Coprocessor | 60% Missing values, conflicts with Graphics | Merged values from Graphics                      | It now uniquely stores the graphics card or coprocessor, consolidation of common information simplifies data retrieval.                                                                             |
| Price                | Outliers                                    | Removed outliers                                 | Outliers exceeding 5x the interquartile range above the upper quartile would significantly bias the data, Figure 3 displays this.                                                                   |
| All                  | Duplicate rows                              | Removed duplicates                               | Duplicate rows are redundant and can skew data.                                                                                                                                                     |

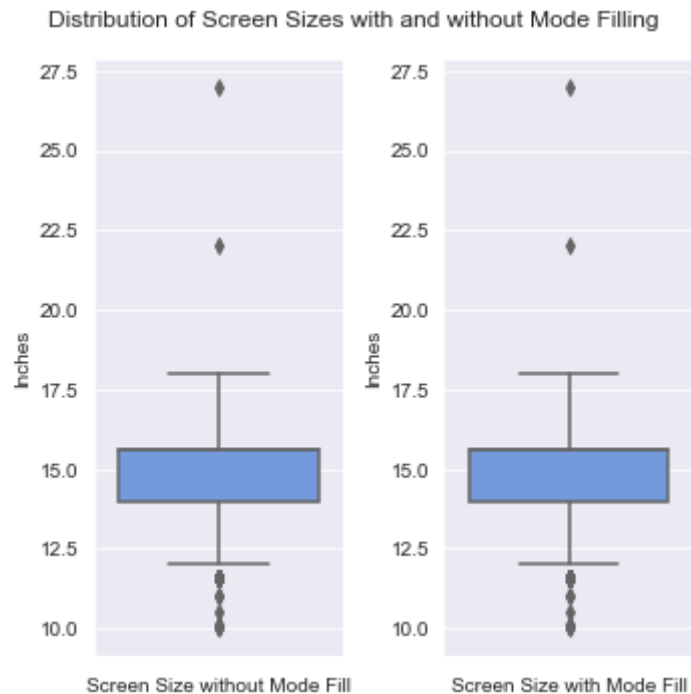


Figure 1 The impact of mode filling on the screen sizes

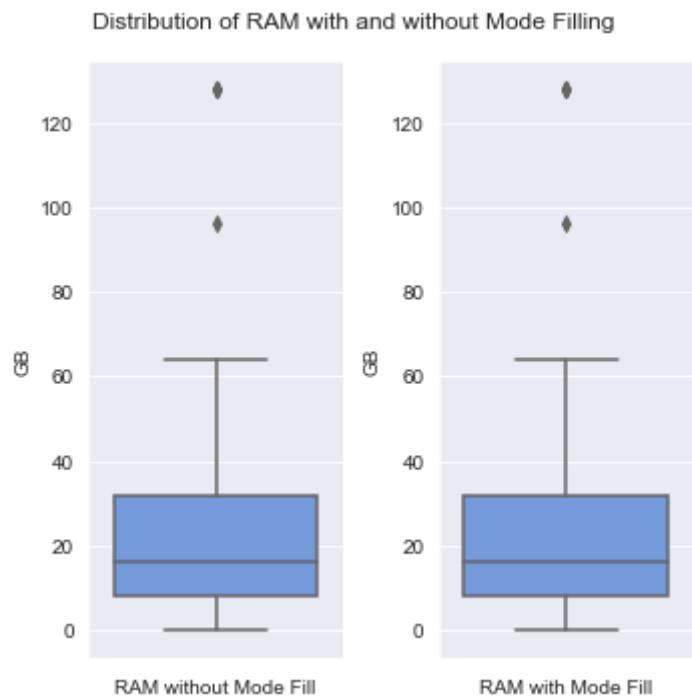


Figure 2 The impact of mode filling on RAM

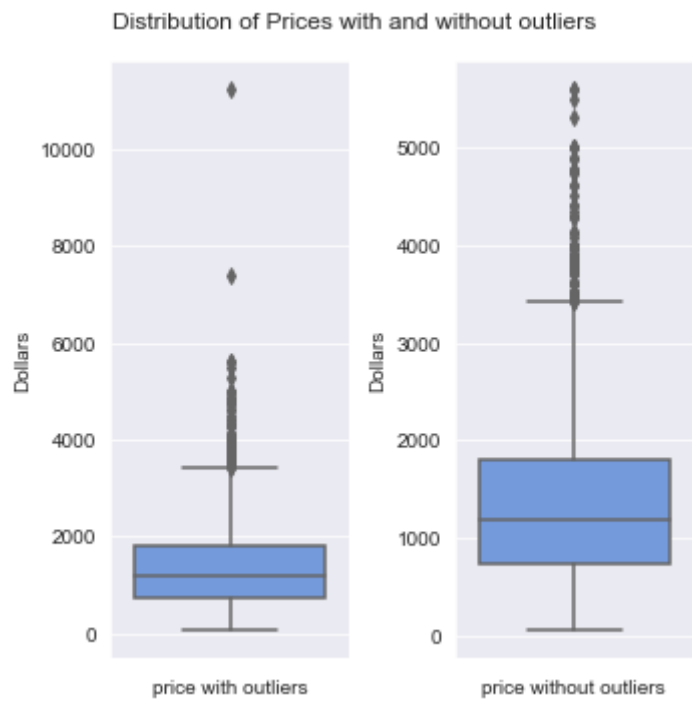


Figure 3 The effect of outlier culling on prices

## Data Analysis

Humanities students likely prefer affordable, portable laptops with high storage that prioritise battery life over processing power, favouring smaller screens and lower processing power. Gamers prioritise processing and graphics capabilities, and larger screens, placing less emphasis on price and storage.

Word Count: 750

Figure 4 suggests there's a moderate positive correlation in specification quality, generally improvements in one component follow improvements in others. While this is optimal, it also implies fewer options for laptops with less processing power but high storage.

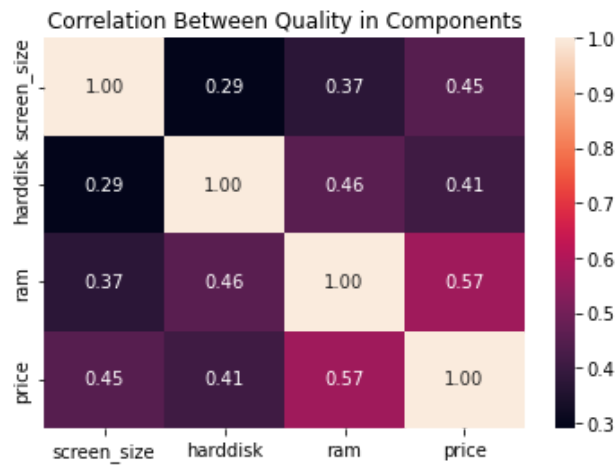


Figure 4 The correlation between numerical components

Larger screens, more RAM, and higher disk space correlate with higher prices and reduced laptop availability. For affordability and portability, screens of  $\leq 14$  inches are optimal averaging \$690. RAM of  $\leq 8$  GB is cost-effective (average \$641), while 1000 GB storage strikes a balance between price (around \$1000) and availability (627). Larger RAM and disk space options have significantly less availability (111) and higher prices (average \$1055).

Word Count: 750



Figure 5 The relationship of components against price

Intel Core i5 processors are the most common (623 laptops) and 86% of laptops have integrated graphics (better battery life). Intel Core i7 closely follows with a higher average price of \$839, however i5 suits lower-end preferences, Celeron, averaging \$305, is a cheaper alternative. Higher-end processors like Ryzen 9 or Core i9 are limited and expensive; therefore, Core i7, with the largest proportion of dedicated graphics (24%), is a potential substitute.

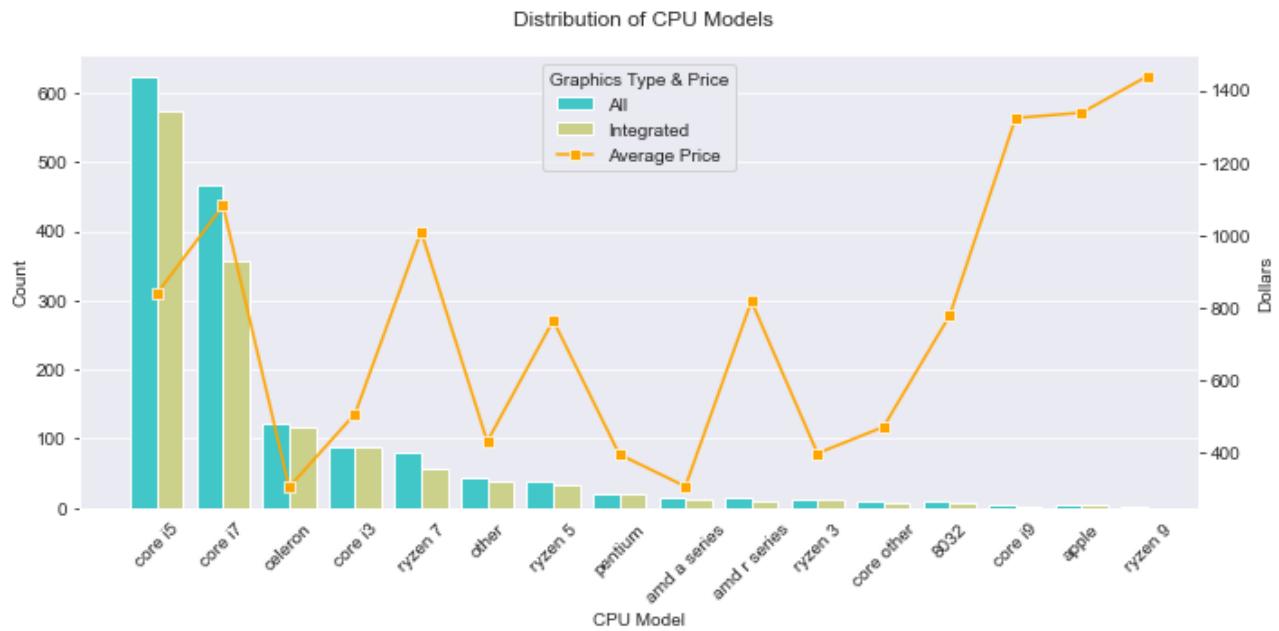


Figure 6 The distribution of CPUs and graphics type

Best laptops for humanities students: **Dell Latitude 3330, Dell Latitude 5320 2-in-1, Dell Latitude 5320, Asus Zenbook S, Dell XPS 9300**. These budget-friendly laptops have low-to-mid-range CPUs, integrated graphics for extended battery life, small screens (~13.5 inches) for portability, and ample storage (~1 TB).

Best laptops for gamers: **Asus Vivobook Pro 15, Shoxlab m15r5, MSI Katana 15, Asus Vivobook Pro 16, MSI Katana 17 b13vfk-835us**. These laptops have high-performance Core i9, Ryzen 9, or Core i7 CPUs, dedicated Nvidia GeForce RTX graphics cards, and 16 or 32 GB RAM.

## References

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